Where does Vera shine?

Daytona sandbox benchmarks on HQ Vera, AMD EPYC 9J45 (Turin) and AWS Graviton5

Status report — data as of 2026-10-14 15:05 UTC. Two of the three hosts are still being measured.

Sources. HQ Vera cell findings (`data/vera-hq-cell-20261014/FINDINGS.md`); phoenix-e6 (Turin) cell findings and live run logs (`/mnt/data/bench/{pin-ab,burst2k,ladder}-20261014_*`); Graviton5 suite logs (`/mnt/data/bench/suite-{4k,64k}-20261014_*`); Daniel's published dataset (`daniel-focus-here/public/`) for the September Vera one-socket ladder and the EPYC 9575F / 9755 points. Every number below was measured with the same Daytona build on all three new cells (runner `353a4060945c` on arm64; the same commit built for x86_64 on Turin), Firecracker v1.16.1, daemon `rlp-rs-0.4.1`, and the same harness and recipe.

1. Summary

Vera leads	**Per-core compute. One job takes 5.9 s on Vera against 6.6–6.7 s on Graviton5 and the EPYC 9575F, and 7.1 s on the EPYC 9755, with almost no contention (c = 44). Packed, Vera delivers about 0.15 jobs/s per physical core (using SMT), against 0.12 on Graviton5 (no SMT), 0.13 on the 9575F and 0.09 on the 9755. The 5.9 s comes from the September one-socket data and still needs re-measuring on the current build.**
Vera leads	**Whole-box throughput so far. 26.6 jobs/s** on HQ Vera (both sockets) against **22.9** on Graviton5 (+16 %). The published 13.3 jobs/s was one socket. Turin has not been measured yet.
No lead	**Platform startup.** With an idle image, 1,000 sandboxes are ready in **≈ 4 s** on Vera and on the 9575F. The earlier 2–3× startup lead came from the benchmark image running a CPU job in every sandbox.
Turin leads	**Startup under load** (agent image, 16 client processes, whole box): Turin **7.9 s**, Graviton5 9.6–15.6 s, Vera 22 s unpinned / 10.2 s pinned to one socket. Spreading over both sockets hurts only Vera.
Turin leads	**Density.** Turin held **4,000** idle sandboxes with 1.6 TB still free; its 5,000 level failed on the API's 60 s create-queue cap, not on the machine. Vera (1.5 TB) and Graviton5 (0.75 TB) are not measured.

Bottom line so far: Vera's advantage is **fast cores under CPU-bound sandbox work** — throughput and per-job latency — not sandbox startup and not density. The Turin throughput ladder is the missing number that decides whether the throughput lead holds against the strongest x86 box.

2. Hosts and setup

	HQ Vera	Turin (OCI phoenix-e6)	Graviton5 (AWS m9gd.metal-48xl)
CPU	NVIDIA Vera, 2 × 88 cores, SMT2 (352 thr)	AMD EPYC 9J45, 2 × 96 cores, SMT2 (384 thr)	Neoverse-V3, 1 socket, 192 cores, no SMT, 2 NUMA nodes
Memory	1,478 GiB	2,267 GiB	754 GiB
Host kernel	6.17 nvidia, **64 KiB pages**, `preempt=none`	6.17 oracle, 4 KiB pages	6.17 aws, 4 KiB (64 KiB A/B running), voluntary preempt
Guest kernel	6.12.34-rlp1 (arm64)	6.8.12-rlp3 (x86_64)	6.12.34-rlp1 (arm64)
Storage	1 NVMe, ext4	1 NVMe, ext4 (12-disk RAID10 removed)	1 NVMe instance store, ext4
"Half" (pinned) layout	socket 0: 88 cores / 176 thr	socket 0: 96 cores / 192 thr	NUMA node 0: 96 cores
Power telemetry	ACPI AP power per socket	RAPL package (no DRAM)	none

Runner settings are identical: 30 s Firecracker socket wait (3212), 512 concurrent VM starts, 2,100 pre-built network namespaces, balloon off, minimum sandbox 0.025 CPU / 64 MiB / 1 vCPU. Only `rlp.slice` is pinned in "half" layouts; API, Postgres, NATS and the client float.

3. Startup: time until all sandboxes are ready

Create-ready burst with probe `echo ready`; "all ready" = burst start ⟶ last sandbox ready and probed. **Idle image** (`alpine:3.20`) measures the platform. The **agent image** (`vera-agent-benchmark:v3`) starts a CPU job of ≈ 3 core-seconds in every sandbox, so it measures startup under load.

3.1. 1,000 sandboxes (seconds)

Image · client · layout	HQ Vera	Turin	Graviton5 4 KiB	Graviton5 64 KiB
Idle · 16 procs · whole box	**4.0 – 4.2**[a]	7.95, 7.96	—	—
Idle · 16 procs · half	**4.0**[a]	—	—	—
Agent · 16 procs · whole box	22.0, 22.5	**7.90, 8.01, 7.97**	9.82, 15.55, 9.56	8.23, 13.71, 11.53
Agent · 16 procs · half	10.2, 10.4	12.52, 11.89, 11.87	15.69, 15.74, 15.83	19.48, 19.48, 16.15
Agent · 1 proc · whole box	20 – 23	10.61 – 11.21	7.07 – 13.79	7.20 – 11.98

Agent · 1 proc · half	8.4 – 10.6	10.22 – 10.80	14.84 – 18.62	18.46 – 18.68

[a] One rep per layout was held open to ≈ 11 s by a single sandbox whose commands hung after it was marked running (3229). EPYC 9575F, same idle test: 3.96 / 3.99 / 3.99 s. All Turin and Graviton5 reps landed every sandbox with zero socket-wait timeouts and zero self-cordons.

3.2. 2,000 sandboxes (seconds)

Image · client · layout	HQ Vera	Turin	Graviton5
Idle · 16 procs · whole box	—	23.22	—
Agent · 16 procs · whole box	27.4 (published Sep, old client)	**22.30, 21.94**	—
Agent · 16 procs · half	—	29.55, 30.97	—

Reading. The platform's own startup is ≈ 4 s per 1,000 on Vera and the 9575F regardless of placement. Turin is the outlier: its idle-image time (≈ 8 s) is the same as its agent-image time, so something other than guest CPU bounds it there — unexplained so far. The single-process client queues requests in Python (accept p50 2.9–5.8 s vs 0.12–0.5 s with 16 processes), so 16-process numbers are the fair ones.

4. Throughput: dense agent ladder (jobs/s)

Published recipe: agent workload, 45 jobs, 8 episodes per sandbox, hold-then-exec, 0.025 CPU (max 1), 64 MiB (max 4 GiB). **c** = sandboxes working at once. Zero failures unless noted.

Host · layout	44	88	176	352	528	704	880	1056	1408	1760	2000
HQ Vera · whole box	—	—	15.19	**26.57**	—	25.89	—	—	—	—	—
HQ Vera · one socket	—	—	14.03	13.11	—	12.96	—	—	—	—	—
Vera · one socket (published Sep)	6.0–6.6	9.5–10.2	11.8–13.7	12.9–13.3	12.9–13.4	12.8–13.1	12.8–13.0	12.8–13.0	12.8–12.9	12.5–12.9[b]	12.7–14.2[b]
Graviton5 4 KiB · whole	5.70	11.94	23.76	22.63	22.44	22.73	22.89	22.79	22.86	22.90	22.96
Graviton5 4 KiB · node 0	5.99	11.94	12.25	12.02	—	11.93	—	11.83	11.85	—	—
Graviton5 64 KiB · whole[c]	5.80	11.91	23.71	23.62	22.97	23.15	—	—	—	—	—
Turin	—	—	—	—	—	—	—	—	—	—	—
EPYC 9575F (published)	5.9	7.4	8.3	8.2	8.2	8.1	8.1	8.0	8.0	8.1	8.1
EPYC 9755 (published)	4.3–5.5	8.3–8.4	9.6–10.0	10.7–11.5	10.6–10.9	10.7–11.3	10.6–11.1	10.7–11.3	10.9–11.4[b]	11.1–13.9[b]	10.9–12.0[b]

[b] Some runs at this level had failures (Vera 2000: 53–627; 9755 from 1408). [c] Running at the time of writing; levels above 704 pending.

4.1. Derived: per-core speed

Host (layout)	Plateau jobs/s	Cores / threads	Jobs/s per core	Jobs/s per thread	Single job p50, c = 44
HQ Vera (whole box)	26.6	176 / 352	**0.151**	0.075	—
Vera (one socket, published)	13.0	88 / 176	**0.148**	0.074	**5.9 s**[d]
Graviton5 4 KiB (whole box)	22.9	192 / 192	0.119	**0.119**	6.65 s
Graviton5 4 KiB (node 0)	11.9	96 / 96	0.124	**0.124**	6.66 s
Graviton5 64 KiB (whole box)	≈ 23.2	192 / 192	0.121	**0.121**	6.63 s
EPYC 9575F (published, 1 socket)	8.1	64 / 128	0.127	0.063	6.72 s
EPYC 9755 (published, 1 socket)	≈ 11	128 / 256	0.086	0.043	7.12 s
Turin	not run	192 / 384	—	—	—

[d] September data on an older build; the guest shape on that host may have allowed 2 vCPUs (each job is still capped at 1 CPU by `--rlp-cpu-max 1`). The gap-fill ladder's c = 44 level re-measures it on the current build.

Vera's lead is per physical core: ≈ 27 % more jobs/s per core than Graviton5 and ≈ 19 % more than the 9575F, and an ≈ 11 % faster single job. Vera gets there with SMT (2 threads per core); per hardware thread Graviton5 is ahead (0.12 vs 0.075). Graviton5 has more cores, which narrows the whole-box gap to 16 %. Platform overhead is 0.5–0.6 % of each job, so these are effectively CPU numbers. Graviton5's 64 KiB host pages change nothing measurable in throughput.

5. Density: how many idle sandboxes fit

Host	3,000	4,000	5,000	Note
Turin	3000/3000 in 42.7 s	4000/4000 in 81.9 s	3941/5000	962 of 1,059 failures were the API's 60 s create-queue cap (`CREATE_MAX_QUEUE_SECS_CAP`); 1,650 GiB free. Wave test running.

HQ Vera	—	—	—	Not measured (needs raised netns/live-VM caps and a /18 subnet).
Graviton5	—	—	—	Not measured.

6. Findings that change how earlier numbers read

- **The startup benchmark measured the image's own workload.** The agent image's entrypoint burns ≈ 3 core-seconds per sandbox; with an idle image Vera and the 9575F both land 1,000 in ≈ 4 s. Startup charts need redoing with an idle image or relabeling as "startup under load".
- **"Control plane accepts the request" is mostly the Python client.** 1 ⟶ 16 processes: accept p50 2.9–5.8 s ⟶ 0.23–0.50 s, all-ready unchanged.
- **NUMA placement only matters on Vera, and only for CPU-heavy guests.** Pinning Vera to one socket helps the agent burst (22 ⟶ 10 s); it halves throughput on Vera and Graviton5 (Turin not yet measured). The "NUMA is the startup lever" claim was withdrawn on 391
- **The 9575F's published startup numbers were inflated** by a background busy loop; its published 2k mean keeps an outlier.
- **Reliability:** 3 s Firecracker socket wait causes spawn timeouts and self-cordons (3212, fixed by the 30 s knob); 2-vCPU snapshot clones oops on Vera ≈ 7 % (3213); the runner marks a sandbox running on a bare TCP accept (3229).
- **Ruled out on Vera:** proxy capacity, event auditing, API restarts, `RLP_VM_MAXACK`, KVM `halt_poll_ns`, the NVIDIA SMT Firecracker fork, memory bandwidth.

7. What is missing, and the plan to fill it

Defined in `data/gapfill-20261014/PLAN.md` with `gapfill.sh`, `gapfill-sampler.py` and `capacity.sh`. Nothing has run; each host waits until its current session is done (one operator per box).

Measurement (one method on all hosts)	HQ Vera	Turin	Graviton5
Startup matrix: idle/agent × whole/half × 1k ×3, 2k ×2, 16 procs	rerun; 2k missing	half + agent 2k missing	idle + 2k missing
Throughput ladder, whole box, c = 44…2000	only 176/352/704	missing	done
Throughput ladder, half, c = 44…1408	only 176/352/704	missing	done
Power and clock during the ladder (jobs/s per watt)	missing	missing	no telemetry
Idle-density ceiling 3,000 ⟶ 6,000	missing	done (4,000+)	missing

Estimated time: ≈ 2.5 h each on Vera and Turin, ≈ 1 h on Graviton5, in parallel. The Graviton5 cell has already rebooted into the 64 KiB kernel (14:19 UTC), so its gap-fill runs on 64 KiB pages unless it is switched back.

8. Caveats

- Host page size differs (Vera 64 KiB; Turin and Graviton5 4 KiB, Graviton5 64 KiB A/B in progress) and so does preemption (Vera `preempt=none`; the AWS kernel cannot switch).
- Guest kernels differ by architecture (x86 6.8.12-rlp3, arm64 6.12.34-rlp1).
- Published September Vera and EPYC data used an older build and, on Vera, likely a forked Firecracker; they are used here only for the per-job and one-socket plateau points.
- Turin's single-NVMe layout deliberately differs from the OCI default RAID10, to match Vera.
- Power scopes will differ (Vera AP module vs AMD package without DRAM); Graviton5 exposes none.